Revisiting Graph Gains: A Reproduction Study of GraphWalker-QA

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Abstract

We reproduce GraphWalker-QA on KaveriHop. With the gold entity annotations used in the original paper we match its results (39.5 versus 39.8 EM on 3-hop questions). With entity links predicted by LinkLite, 3-hop EM drops to 35.1, slightly below HopRank at 35.4. GraphWalker-QA's 3-hop advantage depends on gold entity links; with predicted links it falls below HopRank.

Keywords: reproducibility, graph-based retrieval, entity linking

1 Introduction

GraphWalker-QA [1] reported that walking an entity graph helps most on 3-hop questions. KaveriHop [2] provides gold bridge-entity annotations and asks that they not be used at test time. We noticed that GraphWalker-QA took its entity links from those annotations.

2 Setup

We use the released GraphWalker-QA code and run it twice: with gold links, as in the original paper, and with links predicted by LinkLite [4]. We also re-run HopRank [3] as a dense baseline.

Method	Source	KaveriHop 2-hop EM	KaveriHop 3-hop EM
GW-QA	[1] as reported	50.1	39.8
GW-QA	reproduced, gold links	50.3	39.5
GW-QA	reproduced, predicted links	49.6	35.1
HopRank	reproduced	53.1	35.4

Table 1: Reproduction on KaveriHop (exact match).

3 Results

With gold links we reproduce the original numbers within 0.3 EM. With predicted links the 3-hop score falls by 4.4 EM. Our results contradict the conclusion of the original paper under realistic conditions.

4 Discussion

One of us (J. Weber) co-authored the original paper; the original authors shared their code and confirmed the use of gold links. GraphWalker-QA's 3-hop advantage depends on gold entity links; with predicted links it falls below HopRank. We recommend that graph-based methods always report results with predicted entity links.

References

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Synthetic paper for the GraphMind hackathon dataset. Authors, institutions, datasets and results are fictional.